MILITARY SPECIFICATION SHEET

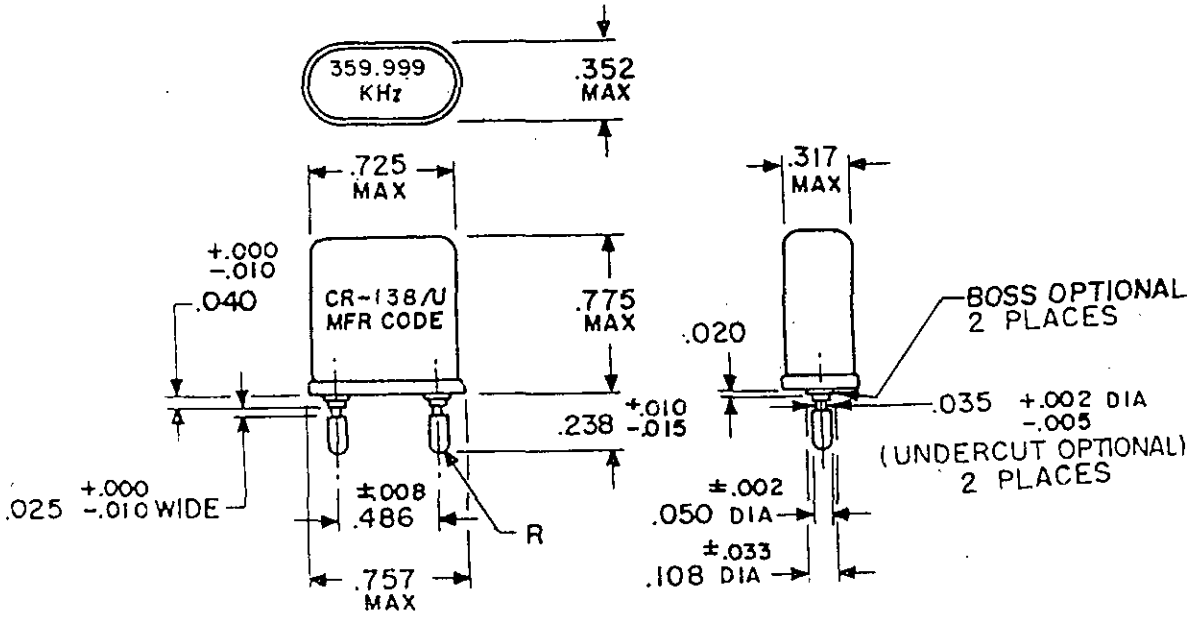
CRYSTAL UNIT, QUARTZ, CR-138/U

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

(B) INACTIVE FOR NEW DESIGN

Pertinent characteristics: 165 to 360 kHz; fundamental; noncontrolled; series.



INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.020	.51	.050	1.27	.352	8.94
.005	.13	.025	.64	.108	2.74	.486	12.34
.008	.20	.033	.84	.238	6.05	.725	18.42
.010	.25	.035	.89	.317	8.05	.757	19.23
.015	.38	.040	1.02			.775	19.69

NOTES:
1. Dimensions are in inches.
2. Metric equivaletns are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. CRYSTAL UNIT, QUARTZ, CR-138/U.

(B) denotes changes

FSC 5955

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-6/U (MIL-H-10056/2)
Dimensions, marking, and configuration	See figure 1
Frequency range	165 to 360 kHz, inclusive
Mode of oscillation	Fundamental
Unwanted modes	Method II, minimum unwanted-to-main mode resistance ratio of 3:1.
Operating temperature ranges:	-40°C to +85°C, inclusive
Frequency tolerance	See table III
Equivalent resistance	See table III
Delta F (Bandwidth) 1/	90 Hertz
Test set, calibration values, and rated drive level	See table II
Shock:	
Frequency change permitted	±10 ppm
Equivalent resistance change permitted	±15 percent
Thermal shock	
Frequency change permitted	±10 ppm
Equivalent resistance change permitted	±15 percent
Vibration: 2/	
Frequency change permitted	±10 ppm
Equivalent resistance change permitted	±15 percent
Bond strength	See table IV

1/ There shall be a difference in frequency of at least 90 Hertz between operation at series resonance and operation at antiresonance into load capacitance of 32.0 ±0.5 pF. The antiresonance measurements shall be made concurrently with the frequency and equivalent resistance measurements made at series resonance over the operating temperature range.

2/ Method 201 of MIL-STD-202.

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	kHz	Ohms	mA	Volt	mW
TS-710/TSM	165 to 180	300	--	0.39	0.5
	180 + to 200	400	--	0.45	0.5
	200 + to 250	300	--	0.39	0.5
	250 + to 360	1000	--	0.71	0.5

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TABLE III. Equivalent resistance and frequency tolerance.

Frequency range inclusive	Maximum resistance	Frequency tolerance
kHz	Ohms	Percent
165 to 199.999	2000	+ 0.000 - 0.024
200 to 360.000 200 to 360.000	3000	+ 0.000 - 0.014

TABLE IV. Bond strength.

kHz	Grams minimum
165 to 250	700
250 + to 320	500
320 + to 360	400

(B) QUALITY ASSURANCE PROVISIONS:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Group A tests of MIL-C-3098 shall be applicable.

NOTE:

1. These crystal units were previously fabricated using Collins Radio Company specification 289-6980-150 thru 289-6980-250.

(B) Custodian:
Air Force - 85

Review activities:
Air Force - 11, 99
DLA - ES

User activities:
Air Force - 17, 19

Preparing activity:
Air Force - 85

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